

Operating system Lab

These are some commands of lab:-

Current directory

```
PS C:\WINDOWS\system32> ubuntu
lab22@DESKTOP-A7A3GTL:~$ pwd
/home/lab22
```

List files and show all files including hidden

```
lab22@DESKTOP-A7A3GTL:~$ ls
lab22@DESKTOP-A7A3GTL:~$ ls -a
.  ..  .bash_history  .bash_logout  .bashrc  .cache  .config  .motd_shown  .profile
```

Create linux Assignment and enter directory

```
lab22@DESKTOP-A7A3GTL:~$ mkdir Linux_Assignment
lab22@DESKTOP-A7A3GTL:~$ cd Linux_Assignment
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ |
```

Create 4 directories

```
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ mkdir documents
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ mkdir programs
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ mkdir reports
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ mkdir backup
```

Display directory structure

```
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ ls
backup  documents  programs  reports
```

Enter Documents

```
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment$ cd documents
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment/documents$ touch file1.txt
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment/documents$ touch file2.txt
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment/documents$ touchfile3.txt
touchfile3.txt: command not found
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment/documents$ touch file.txt
```

Display files

```
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment/documents$ touch file.txt
lab22@DESKTOP-A7A3GTL:~/Linux_Assignment/documents$ ls
file.txt  file1.txt  file2.txt
```

Writing text into file1.txt

```
PS C:\WINDOWS\system32> ubuntu
lab22@DESKTOP-A7A3GTL:~$ echo "Operating System" > file1.txt
lab22@DESKTOP-A7A3GTL:~$ echo "Linux command practice" >> file1.txt
lab22@DESKTOP-A7A3GTL:~$ echo "KRMU" >> file1.txt
lab22@DESKTOP-A7A3GTL:~$ cat file1.txt
Operating System
Linux command practice
KRMU
```

Copy file1.txt to backup, rename, move ,check backup

```

lab22@DESKTOP-A7A3GTL:~$ Linux_Assignment/
-bash: Linux_Assignment/: Is a directory
lab22@DESKTOP-A7A3GTL:~$ Linux_Assignment/Documents
-bash: Linux_Assignment/Documents: No such file or directory
lab22@DESKTOP-A7A3GTL:~$ cp file1.txt ../Backup/
cp: cannot create regular file '../Backup/': Not a directory
lab22@DESKTOP-A7A3GTL:~$ mv file2.txt student.txt
mv: cannot stat 'file2.txt': No such file or directory
lab22@DESKTOP-A7A3GTL:~$ mv file3.tt ../Reports/
mv: cannot stat 'file3.tt': No such file or directory
lab22@DESKTOP-A7A3GTL:~$ ls ../Backup
ls: cannot access '../Backup': No such file or directory
lab22@DESKTOP-A7A3GTL:~$ wc file1.txt
 3  6 45 file1.txt

```

Files information and searching

lines words and character

First few lines

Last few lines

Search for linux

Find all .txt files

```

lab22@DESKTOP-A7A3GTL:~$ wc file1.txt
 3  6 45 file1.txt
lab22@DESKTOP-A7A3GTL:~$ wc file1.txt
 3  6 45 file1.txt
lab22@DESKTOP-A7A3GTL:~$ head file1.txt
Operating System
Linux command practice
KRMU
lab22@DESKTOP-A7A3GTL:~$ grep "Linux" file1.txt
Linux command practice
lab22@DESKTOP-A7A3GTL:~$ cd ..
lab22@DESKTOP-A7A3GTL:/home$ find . -name "*.txt"
./lab22/file1.txt
./lab22/Linux_Assignment/documents/file2.txt
./lab22/Linux_Assignment/documents/file1.txt
./lab22/Linux_Assignment/documents/file.txt
lab22@DESKTOP-A7A3GTL:/home$ |

```

Permissions

Display permissions

Change file1.txt permissions

Verify

```

lab22@DESKTOP-A7A3GTL:/home$ ls -l Documents
ls: cannot access 'Documents': No such file or directory
lab22@DESKTOP-A7A3GTL:/home$ chmod 644 Documents/file1.txt
chmod: cannot access 'Documents/file1.txt': No such file or directory
lab22@DESKTOP-A7A3GTL:/home$ ls -l Documents/file1.txt
ls: cannot access 'Documents/file1.txt': No such file or directory
lab22@DESKTOP-A7A3GTL:/home$ |

```

System information

```

lab22@DESKTOP-A7A3GTL:/home$ uname -a
Linux DESKTOP-A7A3GTL 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
lab22@DESKTOP-A7A3GTL:/home$ whoami
lab22

```

Date

```
Lab22@DESKTOP-A7A3GTL:/home$ date
Thu Oct 1 10:08:35 UTC 2026
```

Df

Filesystem	Size	Used	Avail	Use%	Mounted on
none	3.9G	0	3.9G	0%	/usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none	3.9G	4.0K	3.9G	1%	/mnt/wsl
drivers	476G	74G	403G	16%	/usr/lib/wsl/drivers
/dev/sdd	1007G	1.6G	955G	1%	/
none	3.9G	36K	3.9G	1%	/mnt/wslg
none	3.9G	0	3.9G	0%	/usr/lib/wsl/lib
rootfs	3.8G	2.8M	3.8G	1%	/init
none	3.9G	500K	3.9G	1%	/run
none	3.9G	0	3.9G	0%	/run/lock
none	3.9G	0	3.9G	0%	/run/shm
none	3.9G	80K	3.9G	1%	/mnt/wslg/versions.txt
none	3.9G	80K	3.9G	1%	/mnt/wslg/doc
C:\	476G	74G	403G	16%	/mnt/c
tmpfs	3.9G	0	3.9G	0%	/tmp
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
none	1.0M	0	1.0M	0%	/run/credentials/systemd-resolved.service
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	779M	12K	779M	1%	/run/user/1000

Free -h

```
Lab22@DESKTOP-A7A3GTL:/home$ uptime
10:10:37 up 24 min, 1 user, load average: 0.06, 0.07, 0.02
```

Process Management

```
Lab22@DESKTOP-A7A3GTL:/home$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.1	24324	15388	?	Ss	09:46	0:00	/sbin/init
root	2	0.0	0.0	3180	2204	hvc0	Sl+	09:46	0:00	/init
root	6	0.0	0.0	3180	2044	hvc0	Sl+	09:46	0:00	plan9 --control-socket 7 --log-level 4 --server-fd 8
root	43	0.0	0.2	50400	16644	?	S<s	09:46	0:00	/usr/lib/systemd/systemd-journald
systemd+	75	0.0	0.1	22416	14468	?	Ss	09:46	0:00	/usr/lib/systemd/systemd-resolved
root	84	0.0	0.1	35352	12232	?	Ss	09:46	0:00	/usr/lib/systemd/systemd-udevd
root	104	0.0	0.0	2888	1948	?	Ss	09:46	0:00	/bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -
root	105	0.0	0.0	4472	3096	?	Ss	09:46	0:00	/usr/sbin/cron -f -P
message+	106	0.0	0.0	8820	5392	?	Ss	09:46	0:00	@dbus-daemon --system --address=systemd: --nofork --n
root	110	0.0	0.3	44096	29844	?	Ss	09:46	0:00	/usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
root	127	0.0	0.1	18436	9460	?	Ss	09:46	0:00	/usr/lib/systemd/systemd-logind
syslog	177	0.0	0.0	220548	5796	?	Ssl	09:46	0:00	/usr/sbin/rsyslogd -n -iNONE
_chrony	210	0.0	0.0	20424	6352	?	S	09:46	0:00	/usr/sbin/chronyd -n -F 1 -x
_chrony	220	0.0	0.0	12096	2452	?	S	09:46	0:00	/usr/sbin/chronyd -n -F 1 -x
root	262	0.0	0.0	5492	2764	tty1	Ss+	09:46	0:00	/usr/sbin/agetty --noreset --noclear --issue-file=/et
root	271	0.0	0.4	123460	32704	?	Ssl	09:46	0:00	/usr/bin/python3 /usr/share/unattended-upgrades/unatt
root	317	0.0	0.0	8652	5608	?	Ss	09:46	0:00	login -- lab22
lab22	358	0.0	0.1	22340	13552	?	Ss	09:46	0:00	/usr/lib/systemd/systemd --user
lab22	362	0.0	0.0	22920	4132	?	S	09:46	0:00	(sd-pam)
lab22	389	0.0	0.0	6136	5472	pts/1	Ss+	09:46	0:00	-bash
root	471	0.0	0.0	3188	1116	?	Ss	09:48	0:00	/init
root	472	0.0	0.0	3204	1128	?	S	09:48	0:00	/init
lab22	474	0.0	0.0	6268	5540	pts/0	Ss	09:48	0:00	-bash
lab22	876	0.0	0.0	7160	4360	pts/0	R+	10:11	0:00	ps aux

```
Lab22@DESKTOP-A7A3GTL:/home$ kill 1234
-bash: kill: (1234) - No such process
Lab22@DESKTOP-A7A3GTL:/home$ pgrep sleep
906
Lab22@DESKTOP-A7A3GTL:/home$
```

Redirection & pipes

Create students.txt